An End-to-End Survey Data Analysis Tool for Profiling and Predicting Student Social Media Usage Levels

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An End-to-End Survey Data Analysis Tool for Profiling and Predicting Student Social Media Usage Levels

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ABSTRACT

The retail industry generates large volumes of transactional data containing information about customers, products, sales, quantities, discounts, profits, shipping, and geographical regions. Although such data contains valuable information, conventional reporting systems often focus mainly an historical summartes and do not provide an integrated approach for predictive analysis and customer intelligence. This research presents a Retail Sales Analytics and Customer Intelligence Framework Using Machine Learning and Business Intelligence in transform retail transaction data into meaniigfid and cacinable insights.

The proposed framework integrares data preprocessing, exploratory data analysis, statistical analysis, feature engineering, feature selection, machine learning, customer segmentation, RFM analysis, Customer Lifetime Value analysis, SQL-Boord data processing, scenario analysis, and Power BI visualization. The implemented system uses the Global Superstore dataset containing 51,290 transaction records, 1,590 customers, 32,033 orde3, and 10,299 products. Four regvesion algorithms, mamely Linear Regressior, Decision Tree, Raadion Forest, and XGBoost, are evalaitied for sales prediction. noleg Mean alyeeline Ereor, Root Mean Squared Ereor, and R^2. K-Means clustering is applicd Jor customer segmentation and evaluated using the Silhoutide Score.

The experimental results prove that XGBoost gave an MAE of 32.8337, RMSE of 57.0368, and R^2 rraer of 0.91 l6 in the baseline epperiment: Tha ILSTIE-XGBLSTM hobrial model mait tes temporal features created by LSTM with XGBoost to predict the transaction-leval vales, obtaining an MAE of 32.6091 and RMSE of 53.8745 in the reported hebrid experiment. K-Means clustering gave a Silhoutte Scor of 0.666655 for two clusters. RFM analysis categoritee the customess inin High Valyes, Mediiim Value and Low Value categories, while CLV analysis gives an additional customer value insight. Scenitorie analysis analyses the alwinges in predicted sales under varying discount and quantity situations.

The major contribution of the framework is the integration of predictive analytics and customer intelligence with business intelligence and a proposed Business Recommendation Engiau. The framework provides a compleer pipeline from now retail data to predictive results, customer insights, interactive dashboards, and decision-support information.

Keywords: *Retail Analytics; Customer Intelligence; Sales Prediction; Machine Learning; XGBoost; K-Means; RFM Analysis; Customer Lifetime Value; Power BI; Business Intelligence*

1. INTRODUCTION

1.1 Background

The retail industry has undergone significant transformation due to the increasing availability of digital transaction data. Retail organizations continuously generate information through customer purchases, product transactions, sales, discounts. quantities, profits, shipping records, and regional operations. Such data provides an opportunity to understand customer behavior, evaluate product performance, identify profitable regions, and improve sales planning. However, raw transaction data camott directly provide usefal business information. It must first bo cleaned, transforwed, analyzed, and interpreted.

Traditional retail reporting generally concentrates on descriptive statistics and historical business performance. Reports may provide total sales, total profit, product performance, and regional performance, but these reports often do tno provide predictive capabilitiee. Modom retail analytics requires a combination of descriptive, diagnostic, predictive, and business intelligence techniques.

Machine learning provides an opportunity to extend retail analytics beyond historical reporting. Regression algorithms can be used to estimate sales, while clustering algorithms can identify groups of customers with similar purchasing behavior. RFM analysis provides customer-level measures babed on Receney, Frequency, and Monetary value. Customer Lifetime Value provides another perspective by estimating the historical value associated with individaal customers.

The project therefore proposes an integrated analytics framework rather than a single predictive model. The project roadmap incdnies data collection, preprocessing, EDA, statistical analysis, foature engineering, machine learning, SQL, visualization, Power BI dashboards, and business insights as interconnected stages.

1.2 Problem Statement

Retail datasets contain extensive information about sales, customers, products, profits, discounts, and regions. However, analyzing these variables independently can make it difficult to obtain a complese understanding of retail performance.

Existing analytical approaches may focus primarily on sales forecasting, customer segmentation, RFM analysis, or dashboard visualization. A lack of integratioa betwinee these components can reduce the usefulness of the analytical results for business decision-making.

The problem addressed in this research is therefore:

How can an integrated retail analytics framework combine data preprocessing, exploratory analysis, machine learning, customer segmentation, customer-value analysis, SQL, scenario analysis, and business intelligence to transform retail transaction data into actionable business insights?

1.3 Research Gap

The project review identifies several gaps in existing retail analytics research:

1. *Limited end-to-end integration of Data Analytics, Machine Learning, SQL, and Business Intelligence.*
2. *Limited integration of customer and product intelligence with sales prediction.*
3. *Limited systematic comparison of multiple machine-learning models.*
4. *Limited conversion of analytical results into actionable recommendations.*
5. *Limited use of Explainable AI for interpreting machine-learning-based retail insights.*

The proposed framework addresses these gaps by integrating multiple analytical components into one workflow.

1.4 Research Objectives

1. The objectives of the proposed research are:
2. To develop an end-to-end retail analytics framework integrating Python, data preprocessing, EDA, machine learning, SQL, and Power BI.
3. To analyze sales, customer, product, profit, discount, and regional behavior.
4. To develop and compare multiple machine-learning models for sales prediction.
5. To perform customer segmentation using K-Means clustering.
6. To perform RFM-based customer-value analysis.
7. To calculate Customer Lifetime Value using historical customer behavior.
8. To develop interactive Power BI dashboards for retail business analysis.
9. To perform model-based scenario analysis for discount and quantity variations.
10. To design a Business Recommendation Engine that converts analytical outputs into business-oriented recommendations.
11. The project documentation identifies the same core objectives, including end-to-end analytics, customer and product behavior analysis, machine-learning evaluation, dashboards, and a business recommendation module.
12. To develop and evaluate a proposed hybrid TATFF-XGBLSTM model that integrates temporal sales patterns learned by LSTM with transaction-level features processed by XGBoost for improved retail sales prediction.

1.5 Research Questions

- **RQ1:** How do Linear Regression, Decision Tree, Random Forest, and XGBoost perform for retail sales prediction?
- **RQ2:** What number of clusters provides the most appropriate customer segmentation according to the Silhouette Score?
- **RQ3:** How does RFM analysis complement machine-learning-based customer segmentation?
- **RQ4:** How can Customer Lifetime Value provide additional information about customer value?
- **RQ5:** How can interactive Power BI dashboards transform analytical results into business intelligence?
- **RQ6:** How can scenario analysis connect machine-learning predictions with business-oriented decision support?

1.6 Contributions

- Development of an integrated retail analytics architecture.
- Comparative evaluation of four sales-prediction algorithms.
- Customer segmentation using K-Means.
- RFM-based customer-value analysis.
- Customer Lifetime Value analysis.
- SQL-based data storage and querying.
- Interactive Power BI business intelligence dashboards.
- Model-based scenario analysis.
- Proposed Business Recommendation Engine
- The study proposes a hybrid TATFF-XGBLSTM model that integrates LSTM-based temporal features with XGBoost to improve retail sales prediction..

1.7 Organization of the Paper

The first section explains the research background, problem, gap, objectives, research questions, and contributions of the research. The second section provides an overview of the research related to retail analytics, machine learning, customer segmentation, RFM analysis, and business intelligence. The third section explains the research methodology and system architecture. The fourth section explains the experimental setup, dataset, tools used, modeling technique, validation process, and performance metrics. The fifth section presents the results of the experiments through tables, graphs, customer analysis, scenario analysis, and the output of the Power BI dashboard. The sixth section discusses the results of the experiment and the practical applications thereof. The seventh section discusses the limitations of the research.

2. RELATED WORK

2.1 Existing Approaches

Typically, retail analytics includes descriptive statistics, sales consolidation, customer analysis, product analysis, and

geographical analysis. Using such approaches, companies can learn about their past performance in business.

But, descriptive analytics is not sufficient for predicting the future performance of sales or customers. Machine learning can help to continue the analytics process by finding nonlinear dependencies within the transactional data.

2.2 Machine Learning / Computational Approaches

Regression algorithms are widely applied in numerical forecasting problems. Linear Regression serves as a simple starting point to study linear dependency between input variables and sales. Decision Trees apply nonlinear decision-making procedures. The Random Forest method applies several decision trees for variance reduction, while XGBoost employs gradient boosting in order to create an ensemble of decision trees.

The suggested research studies the performance of all four approaches on the same dataset and the same test split.

2.3 Advanced / Alternative Approaches

Customer Segmentation involves segregating the customers on the basis of similarities in purchase behavior. The use of clustering algorithms is beneficial because they don't need pre-existing customer labels.

The K-Means algorithm is probably the most popular clustering technique. Customers are segregated on the basis of closeness to the cluster centroids.

The RFM model offers another customer segmentation technique. RFM stands for:

Recency – How recent was the last purchase of the customer?

Frequency – How often did the customer make purchases?

Monetary – How much money did the customer spend?

There have been many studies regarding the use of machine learning and clustering techniques in retail customer segmentation. John, Shobayo, and Ogunleye compared different clustering algorithms through RFM customer features and found differences in clustering effectiveness.

Wang, Sun, and Yu developed a dynamic customer segmentation model by combining LRFMS with multivariate time-series clustering. It highlights the significance of modeling customer behavior dynamicalle.

2.4 Explainability / Interpretability (or other relevant sub-theme)

Kasem, Hamada, and Taj-Eddin conducted research in customer profiling, segmentation, and sales prediction through artificial intelligence in direct marketing. The research shows how useful it is to integrate the use of customer profiling and prediction in marketing analytics.

Tang studied retail demand forecasting and customer classification through regression techniques and RFM analysis. This gives an example of how sales prediction and customer segmentation can be integrated in retail analytics.

It can therefore be seen that while it is possible to apply machine learning and customer analytics in retail, different researches tend to focus on different parts of the whole analytics process.

2.5 Critical Comparison

The study by Kasem, Hamada, and Taj-Eddin involves customer profiling, segmentation, and sales predictions using artificial intelligence in direct marketing. The study is an example of the importance of combining the use of customer profiling and sales prediction in marketing analytics.

Tang researched retail demand forecasting and customer classification using regression methods and RFM analysis. This study provides an example of the integration of sales prediction and customer segmentation in retail analytics.

It can therefore be deduced that although machine learning and customer analytics are applicable in retail, different studies have concentrated on different aspects of the whole analytics process..

#	Study / Publication	Key Contribution	Limitation
1	Ulocking Retail Insights: Predictive Modeling and Customer Segmentation Through Data Analytics	Predictive modeling and customer segmentation	Limited BI, SQL, and actionable recommendation integration
2	Predictive Modeling and Customer Segmentation Using ML	ML model comparison for sales forecasting	Limited customer intelligence and BI integration
3	Customer Profiling, Segmentation, and Sales Prediction Using AI in Direct Marketing	Customer profiling, segmentation and sales prediction	Limited product, regional and BI integration
4	A Dynamic Customer Segmentation Approach by Combining LRFMS and	Dynamic customer segmentation	Limited sales prediction and broader BI integration

#	Study / Publication	Key Contribution	Limitation
	Multivariate Time Series Clustering]		
5	A Comprehensive Framework for Customer Segmentation and Predictive Analytics	Customer segmentation and predictive analytics	Limited SQL, Power BI and automated recommendations

2.6 Research Gap (Summary)

The reviewed literature shows the importance of using machine learning, clustering, RFM analysis, and customer profiling. But there is still a need for an integrated approach that includes sales prediction, customer segmentation, customer value analysis, SQL, visualization, and business insights.

The suggested approach satisfies the need through creating an analytical pipeline:

Data → Data Preprocessing → Exploratory Data Analysis → Feature Engineering → Machine Learning → Customer Intelligence → SQL → Power BI → Scenario Analysis → Business Insights.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

proposed methodology follows an end-to-end architecture that converts raw retail transactions into analytical and business intelligence outputs..

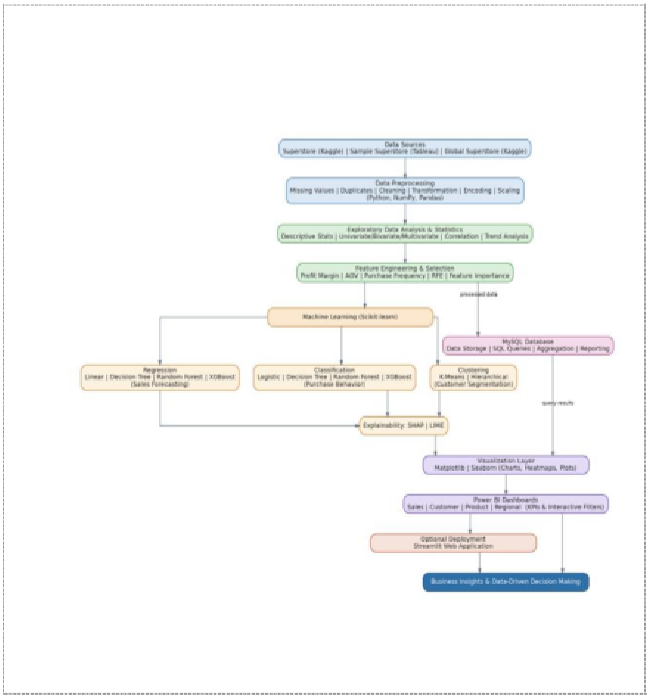


Fig. 1.The roadmap provided for the project includes an integrated pipeline process starting from data sources to visualization using Power BI and MySQL along with various other steps such as machine learning..

Proposed Hybrid Model: TATFF-XGBLSTM:

This research proposes TATFF-XGBLSTM, a Temporal-Aware Transaction Feature Fusion model that integrates Long Short-Term Memory (LSTM) networks with Extreme Gradient Boosting (XGBoost) for retail sales prediction. The model combines historical temporal information with transaction-level attributes to improve the representation of retail sales behavior.

The proposed architecture consists of three principal stages: temporal feature generation, feature-level fusion, and final sales prediction. First, transaction-level sales data from the Global Superstore dataset are aggregated into monthly sales totals. The LSTM model receives the previous six months of sales history and generates a one-step temporal prediction for the subsequent month.

In the second stage, the generated temporal prediction is incorporated into the corresponding month's transaction records as an additional input feature. The resulting dataset contains 22 transaction-level predictors and one LSTM-generated temporal feature. This feature-fusion strategy allows the model to combine historical sales patterns with product, geographic, shipping, and other transaction attributes.

In the final stage, XGBoost processes the combined features and predicts transaction-level sales. XGBoost is used as the final predictor because it captures nonlinear relationships among structured transaction attributes. The LSTM component supplies additional temporal context rather than acting as the final sales predictor.

The proposed model is evaluated against standalone machine-learning models, standalone deep-learning models, and alternative hybrid configurations. The evaluation uses Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), Mean Absolute Percentage Error (MAPE), and Symmetric Mean Absolute Percentage Error (SMAPE), where applicable.

To preserve temporal ordering, the experiments use a chronological training-test split. The temporal feature is generated using historical monthly sales information, with a six-month lookback. The study also acknowledges limitations associated with the relatively short 48-month series, the single hold-out evaluation, and the possibility of information leakage from certain transaction attributes.

3.2 Dataset(s)

The suggested approach takes into consideration publicly accessible Superstore retail databases as reference databases. The road map for the project had outlined the availability of Superstore Sales Data Set, Sample Superstore Data Set, and Global Superstore Data Set as potential sources of data. In the experiment conducted for the final results that are presented in this paper, Global Superstore Data Set has been chosen.

- Dataset 1 — The Global Superstore Dataset was sourced from Kaggle and consists of transactions of a retail store in various locations. The adopted dataset has 51,290 observations and 24 variables. It has data about 1,590 customers, 25,035 orders, and 10,292 items and the date range is between 2011 and 2014 The dataset was selected because it provides sufficient diversity in customer, product, sales, profit, discount, quantity, order, and regional attributes for performing both predictive analytics and customer intelligence analysis. The dataset also supports the development of machine-learning models, customer segmentation, RFM analysis, Customer Lifetime Value analysis, SQL queries, and Power BI dashboards.
- The major variables used in the analysis include:
-
- Product Name
- Category
- Sub-Category
- Sales
- Quantity
- Discount
- Profit
- Shipping information

  Dataset 2 —The data set was chosen based on the following facts:

- The data set contains enough diversity of customer, product, sales, profit, discount, quantity, order, and region-related attributes to perform predictive analytics and customer intelligence. The data set is suitable for creating machine learning algorithms, customer segmentation, RFM analysis, CLV analysis, SQL queries, and Power BI reports.
- The key variables considered during the analysis were:
- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Country
- City
- State
- Region
- Product ID
- Product Name
- Category
- Sub-Category
- Sales
- Quantity
- Discount
- Profit
- Shipping information.
-

3.3 Data Preprocessing

Data pre-processing was done in order to convert the raw retail transaction data into a consistent form that would be useful for exploration, feature extraction, machine learning, customer segmentation, and visualization.

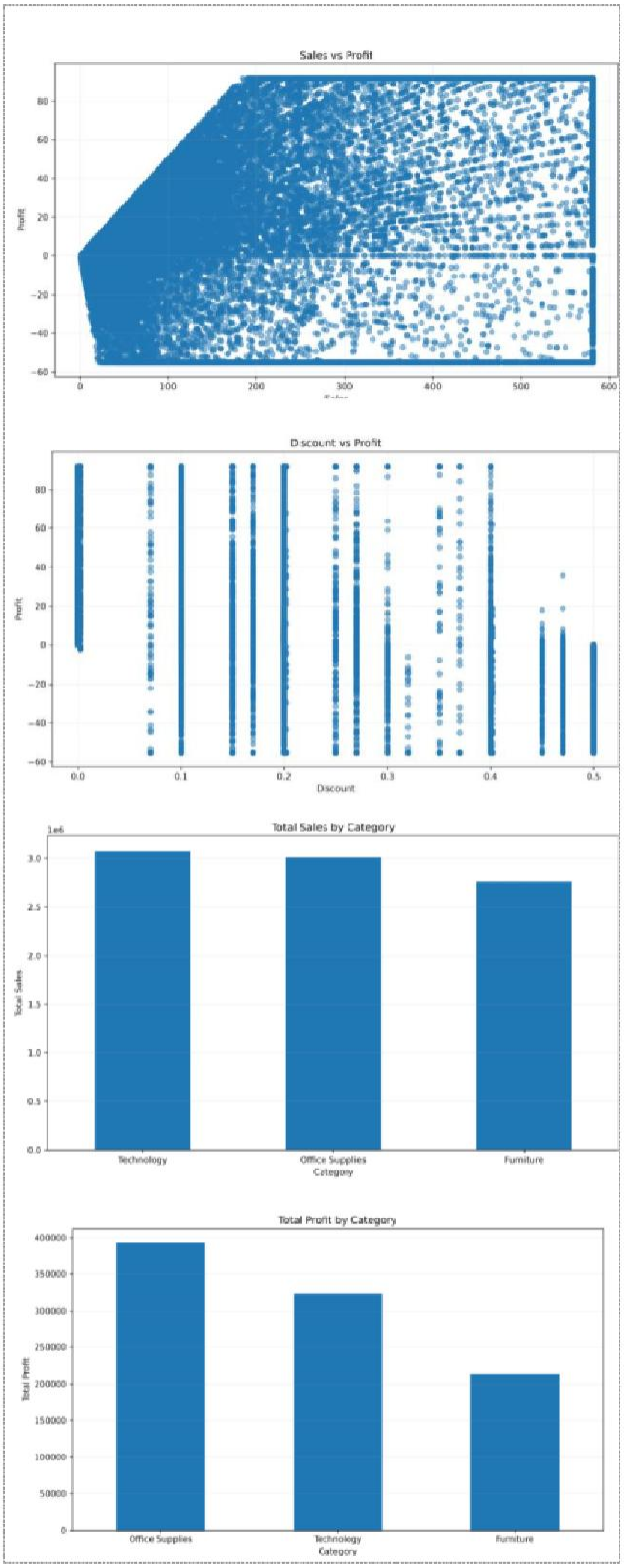
The pre-processing pipeline involved the following steps.

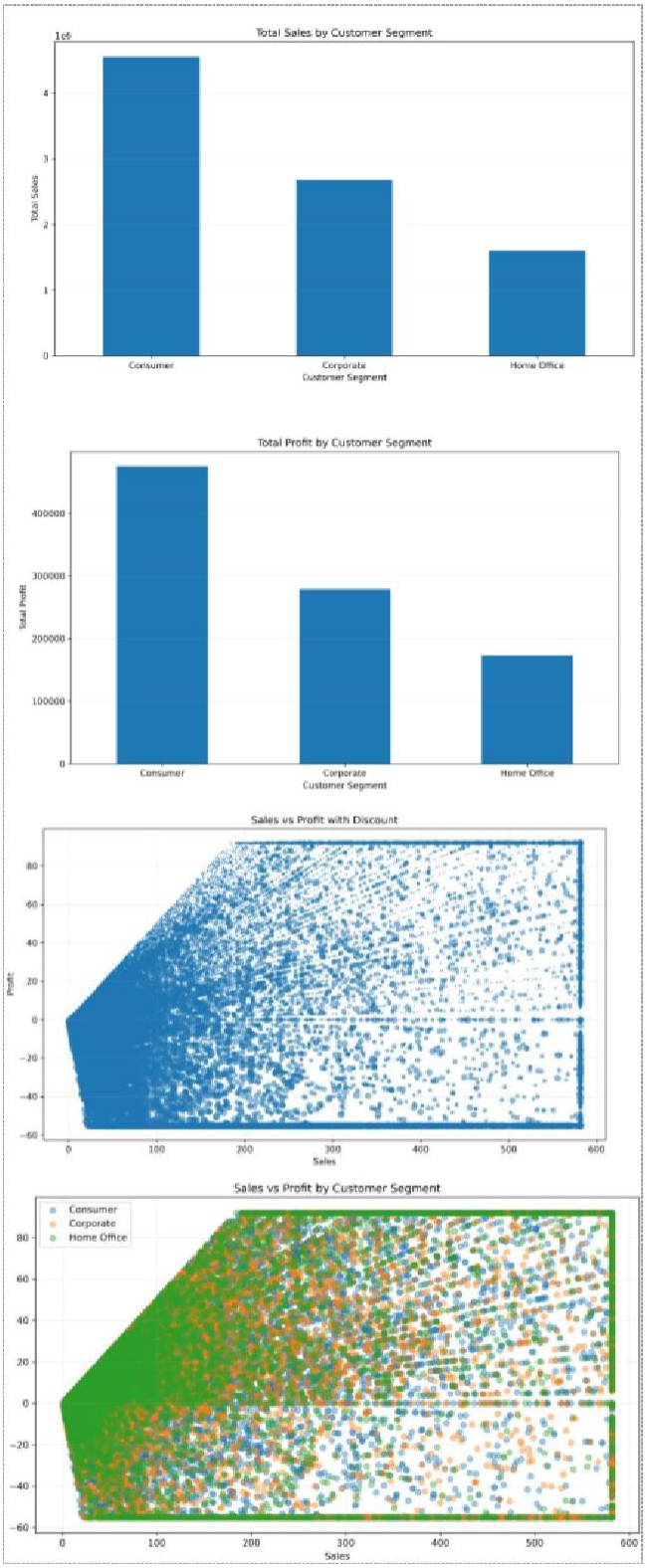
- Data inspection helped ascertain the dimension of data, name of attributes, type of data, numerical variables, categorical variables, and date variables.
- There were 51,290 observations and 24 columns in the dataset.
- Step 2 – Missing Values Detection
- The presence of missing value in the data was analyzed. Those columns which had missing values were detected and dealt with based on their analytical significance prior to building the model.
- This was required since missing values would impact statistical analysis, creation of features, clustering, and prediction by the machine learning algorithm.
- Step 3 – Duplicate Records
- Duplicate records were analyzed and eliminated wherever needed to avoid counting repeated transactions for generating statistical information and training of the model.
- Step 4 – Data Type Conversion
- The attributes like Order Date and Ship Date were changed to date format which allowed the creation of year, month, quarter and duration of shipment features.
- Step 5 – Outlier Detection

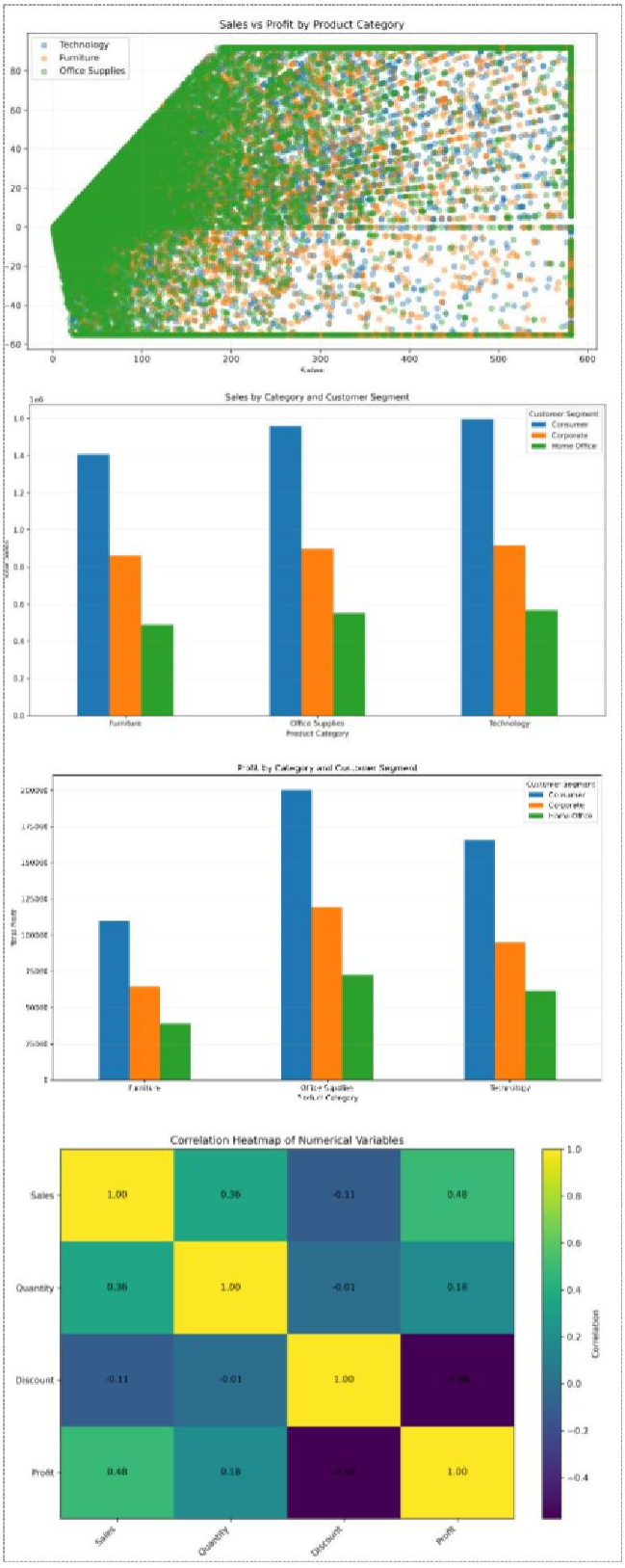
- Numeric attributes like Sales, Quantity, Discount, and Profit were checked for any unusual observation. Outlier detection was used in the Exploratory Analysis phase to know the distribution of retail transactions.
- Step 6 – Categorical Data Handling
- Categorical variables were encoded to machine learning compatible format wherever necessary. The attributes that were used in prediction models were suitably encoded.
- 7 – Feature Scaling
- Customer-level features utilized for clustering were standardized such that variables having higher numerical values did not overshadow distance calculation.
- Step 8 – Data Transformation
- Further transformations were conducted to transform transaction-level data to customer-level and product-level analytical data.
- Preprocessing therefore resulted in an organized analytical data set that could be used uniformly for EDA, machine learning, customer segmentation, SQL, and Power BI.

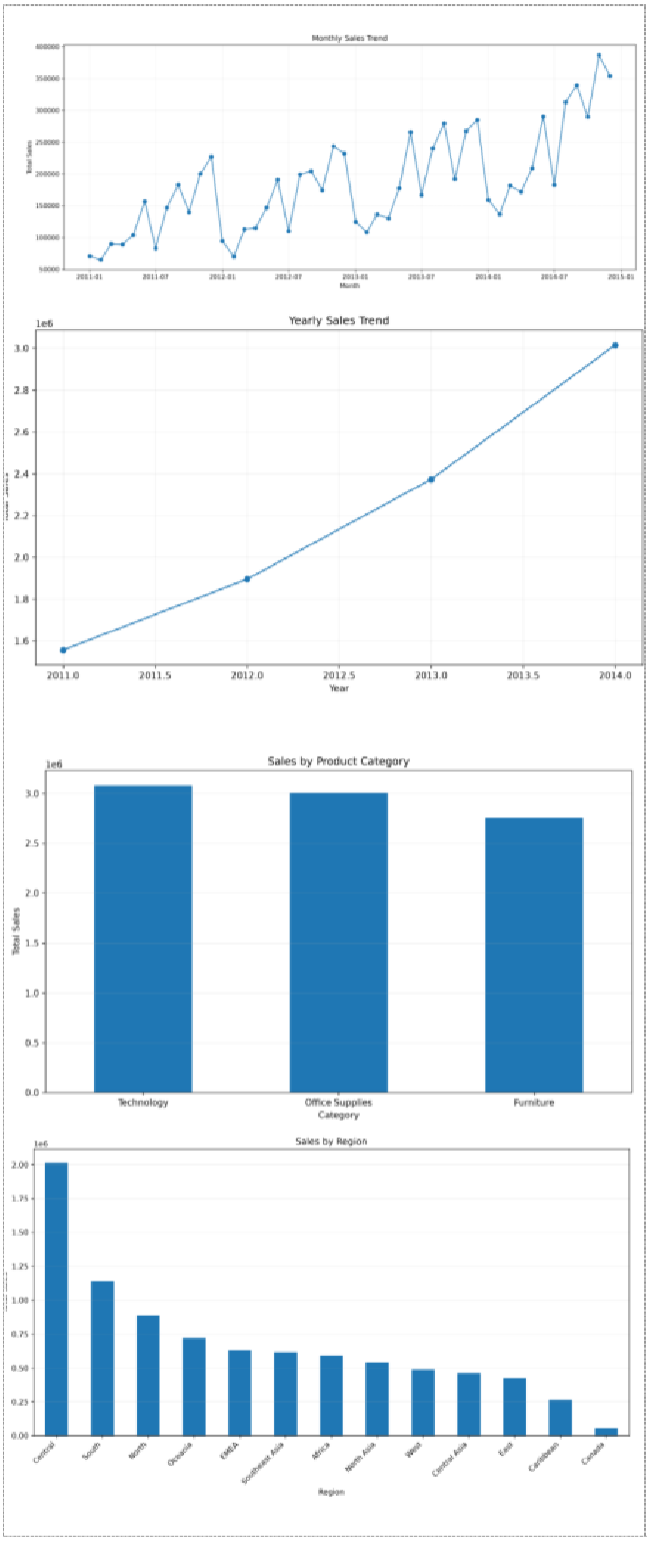
3.4 [Exploratory Analysis / Feature Engineering, as relevant]

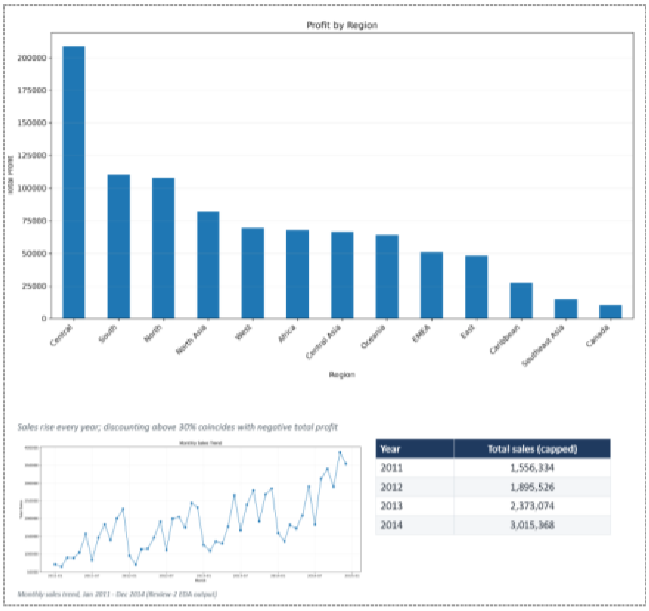
Exploratory Data Analysis (EDA) was performed to understand the structure, distribution, relationships, and business characteristics of the retail dataset before applying machine-learning models.




Total Sales by Customer Segment
1e6
Total Sales
Consumer
Corporate
Home Office
Customer Segment
Total Profit by Customer Segment
Total Profit
Consumer
Corporate
Home Office
Customer Segment
Sales vs Profit with Discount
Profit
Sales
Sales vs Profit by Customer Segment
Consumer
Corporate
Home Office
Sales


Sales vs Profit by Product Category
Technology
Furniture
Office Supplies
Sales by Category and Customer Segment
1e6
Customer Segment
Consumer
Corporate
Home Office
Furniture
Office Supplies
Technology
Product Category
Profit by Category and Customer Segment
Customer Segment
Consumer
Corporate
Home Office
Correlation Heatmap of Numerical Variables
Sales
Quantity
Discount
Profit
1.00
0.36
-0.11
0.48
0.36
1.00
-0.01
0.18
-0.11
-0.01
1.00
-0.58
0.48
0.18
-0.58
1.00
Correlation





Proposed Hybrid Model: TATFF-XGBLSTM:

- Objective: The proposed hybrid model leverages temporal patterns in sales and transactional attributes to enhance sales predictions at retail.
- LSTM component: Utilizes the last six months' sales data to make a temporal prediction for the next month.
- Feature combination: The temporal prediction by LSTM is added to the transaction level dataset as one more attribute.
- XGBoost component: Processes the temporal and transaction attributes to make a sales prediction.
- Hybrid Model Evaluation:** The hybrid model is assessed in comparison with some base line models, using evaluation metrics like MAE, RMSE, R^2, MAPE, and so forth.
- -Expected Outcome:** The hybrid model is expected to help sales forecasting by combining temporal sales trends with nonlinear relations among attributes.

	XGBoost	Random Forest
LSTM	LSTM + XGBoost (proposed basis)	LSTM + Random Forest
GRU	GRU + XGBoost	GRU + Random Forest

3.5 Proposed Model(s) and Baselines

Four regression algorithms were implemented for sales prediction. A baseline model and three tree-based approaches were selected to compare linear and nonlinear learning capabilities.

- Linear Regression was included as the baseline regression algorithm. The algorithm uses a linear function to model the relationship between the input and the output. The simplicity of the model is what makes it useful as a benchmark.
- Decision Tree Regressor
- The reason for including Decision Tree Regression algorithm is that it is capable of modeling nonlinear relations with decision rules. Decision tree algorithms are specifically suitable for heterogeneous data sets.
- Random Forest Regressor
- The technique of Random Forest Regression involves building the predictions from several decision trees. The process may lower the variance of the model in comparison with using only one decision tree. Breiman developed Random Forests as an approach to ensemble learning with the use of randomized decision trees. [1]
- XGBoost Regressor
- The algorithm XGBoost was selected among gradient boosting models. It builds decision trees in a sequential manner to correct the errors made by the previous trees. XGBoost was invented by Chen and Guestrin
- Proposed Hybrid Model: TATFF-XGBLSTMTATFF-XGBLSTM is proposed by this research work which stands for Temporal-Aware Transaction Feature Fusion model. This model combines LSTM network along with XGBoost model for predicting retail sales. The proposed model utilizes the past temporal data along with transaction features for enhancing the representation of retail sales behavior.

The overall architecture is comprised of three main stages: temporal feature creation, feature-level fusion, and sales prediction. Initially, the transaction-level sales data from the Global Superstore dataset is aggregated to create monthly sales sums. The LSTM takes the last six months of sales data and creates a one-step-ahead temporal prediction of the next month.

In the second step, the temporal feature created by the model is then fused with the corresponding month's transaction-level data by adding it as an additional feature. Therefore, the new dataset has 22 transaction-level features and 1 temporal feature produced by the LSTM.

In the final step, XGBoost uses the fused features to predict the transaction-level sales. XGBoost is selected as the final predictor because it takes into account the non-linear relationships between transaction attributes. The role of the LSTM model is to add temporal context to the predictions rather than predicting sales directly.

The proposed model is compared to standalone machine learning models, standalone deep learning models, and other possible hybrid models. The comparison is done based on MAE, MSE, RMSE, coefficient of determination (R^2), MAPE, and SMAPE, where appropriate. To maintain the time ordering, the experiments will employ chronological training and testing data.

The temporal attribute will be created through the use of historical monthly sales data, looking back by six months. Some of the limitations considered in this study include the relatively short 48-month sequence, one-time evaluation of the hold out data, and potential information leakages from some transactional features.

3.6 Algorithmic / Mathematical Formulation

3.6.1 Linear Regression

The Linear Regression model can be represented as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \qquad (1)$$

where:

- $\hat{y}$ *represents predicted sales.*
- β_0 *represents the intercept.*
- β_i *represents the coefficient associated with feature* x_i.
- p *represents the number of input features.*

3.6.2 Mean Squared Error Objective

The regression models minimize prediction error using loss functions related to the difference between actual and predicted values.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \qquad (2)$$

where y_i *represents the actual sales value and* $\hat{y}_i$ *represents the predicted sales value.*

3.6.3 Random Forest Prediction

For an ensemble containing B regression trees, the Random Forest prediction can be represented as:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(x) \qquad (3)$$

where $T_b(x)$ is the prediction of the bth decision tree.

3.6.4 K-Means Objective Function

K-Means clustering minimizes the within-cluster sum of squared distances:

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \| x_i - \mu_k \|^2 \qquad (4)$$

where:

- *K is the number of clusters.*
- *C_k is cluster k.*
- *x_i is a customer observation.*
- *μ_k is the centroid of cluster k.*

3.6.5 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^{n} | y_i - \hat{y}_i | \qquad (5)$$

3.6.6 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \qquad (6)$$

3.6.7 Coefficient of Determination

R2=1−∑i=1n(yi−y¯)2∑i=1n(yi−yi^)2

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

There is no need for specialized hardware, and the system will run on a standard PC.

Software:

The following software was used:

Python

NumPy

Pandas

Matplotlib

Seaborn

Scikit-learn

XGBoost

MySQL

MySQL Workbench

Microsoft Power BI

Jupyter Notebook/Visual Studio Code

Git and GitHub

According to the project roadmap, the following project technologies were identified: Python, Jupyter/VS Code, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, MySQL, MySQL Workbench, Power BI, Git, and GitHub.

Hardware:

It was sufficient to use a standard PC with enough memory and storage to process 51,290 records of transactions. There is no need for GPU or any other specialized hardware for conducting experiments.

4.2 Dataset Split / Design

The number of data records in the Global Superstore database is 51,290.

The split ratio for the regression tests was 80:20.

Dataset Portion	*Number of Records*
Complete Dataset	*51,290*
Training Dataset	*41,032*
Testing Dataset	*10,258*

The training dataset was utilized for regression model fitting, whereas the testing dataset was used for performance evaluation of the model.

The same test dataset was used for comparison between Linear Regression, Decision Tree, Random Forest, and XGBoost under the current experimental setup.

The customer segmentation process started with creation of customer level features and clustering using K-Means method. Different cluster numbers from K = 2 to K = 6 were tested based on the Silhouette Score.

4.3 Hyperparameter Configuration

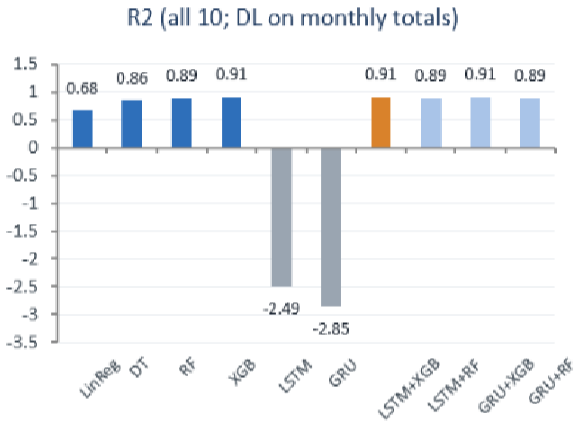
The regression models were implemented using the configurations used in the project experiment.

Model	***Main Configuration***
Linear Regression	*Standard Linear Regression*

Model	*Main Configuration*
Decision Tree	*Maximum depth = 12*
Random Forest	*150 estimators; maximum depth = 15*
XGBoost	*300 estimators; learning rate = 0.05; maximum depth = 6; subsample = 0.8; column sampling = 0.8*
K-Means	*K values from 2 to 6 evaluated*

The configurations were selected to provide a comparison between a simple linear baseline, an individual decision tree, an ensemble of trees, and a gradient-boosting approach.

For K-Means, the number of clusters was not fixed in advance. Instead, multiple values of K were evaluated using the Silhouette Score



4.4 Validation Strategy

The regression models were validated by means of the 80:20 hold-out validation strategy.

The training dataset included 41,032 observations, and the testing dataset included 10,258 observations.

The performance of models was assessed based on the test dataset using such metrics as MAE, RMSE, and R².

In case of customer segmentation, K-Means model with cluster configuration varying from two to six clusters was validated using the Silhouette Score metric.

Neither repeated k-fold cross-validation nor bootstrap validation was employed in this experiment. Hence, the results presented are obtained in the context of hold-out test set only.]

4.5 Evaluation Metrics

4.5.1 Mean Absolute Error

- Mean Absolute Error measures the average absolute difference between actual and predicted sales.
- $MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$
- A **lower MAE** indicates smaller average prediction errors.
- MAE was selected because it provides an easily interpretable measure of the typical prediction error.

4.5.2 Root Mean Squared Error

- RMSE measures the square root of the average squared prediction error.
- $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$
- A **lower RMSE** indicates better predictive performance.
- RMSE was included because squaring the errors gives greater influence to larger prediction errors, which is relevant when unusually large sales-prediction errors need to be detected.

4.5.3 R² Score

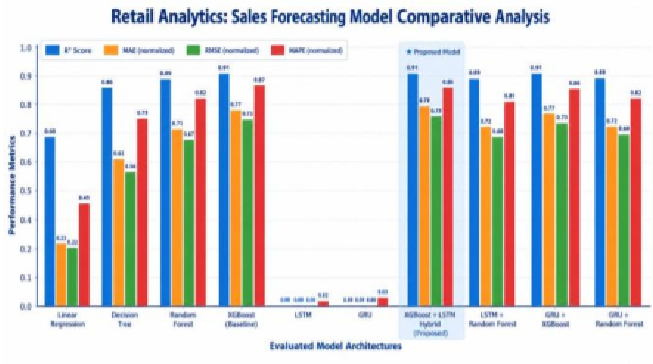
- R^2 measures the proportion of variance in the target variable explained by the model.
- $R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$
- A **higher R²** indicates that the model explains a larger proportion of variation in sales.
- R² was selected to provide a standardized measure for comparing the four regression models.

4.5.4 Silhouette Score

- The Silhouette Score measures the separation and compactness of clusters.
- For an observation i:
- $$s(i) = \frac{b(i) - a(i)}{\max\left(a(i), b(i)\right)} \qquad (8)$$
- where:
- $a(i)$ represents the average distance between observation i and other observations in the same cluster.
- $b(i)$ represents the lowest average distance between observation i and observations belonging to another cluster.
- A **higher Silhouette Score** indicates better-defined clusters.
- The metric was selected because customer segmentation requires both high within-cluster similarity and good separation between clusters.

5. RESULTS

This section contains the findings that have been derived empirically through the Global Superstore experiment that has been conducted. The findings contain the performance of the sales prediction, customer segmentation, RFM analysis, CLV, correlation analysis, scenario analysis, and Power BI dashboards.



5.1 Overall Performance

performance achieved by Linear Regression was an R^2 of 0.6830. On the other hand, Decision Tree and Random Forest had R^2 performances of 0.8682 and 0.8902, respectively. However, XGBoost produced the best R^2 of 0.9116, along with the best MAE of 32.8537 and the best RMSE of 57.0368.

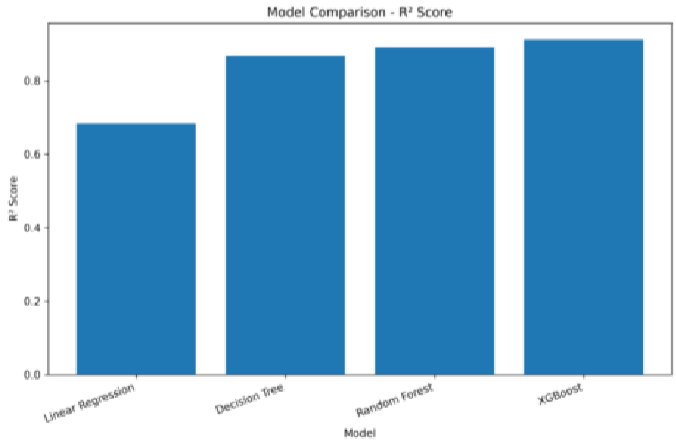
These outcomes show that the tree-based algorithms performed better in identifying nonlinear patterns in the retail transaction data than the linear algorithm under the experimental framework used.

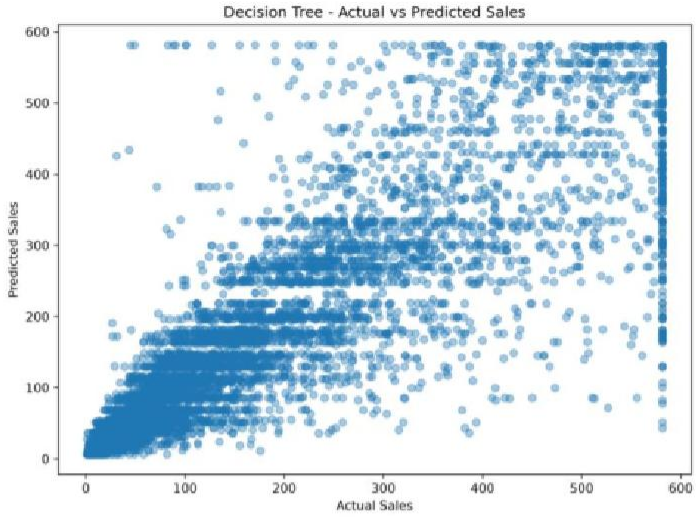
5.2 Model / Method Comparison

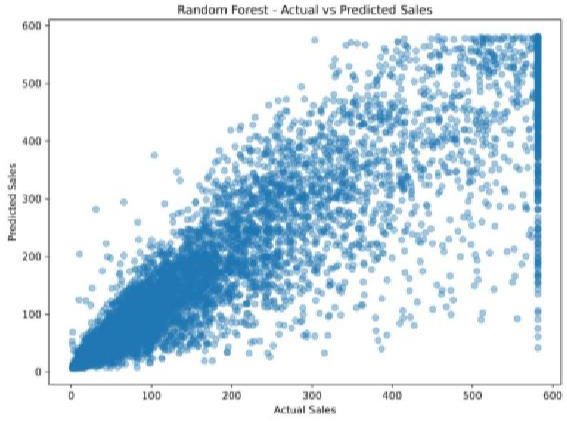
Table 2 compares the performance of regression models used for retail sales and prediction .

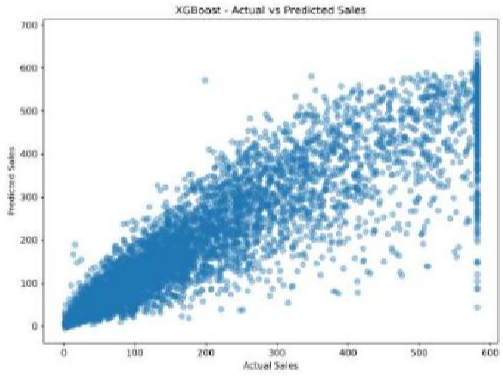
Method	MAE	RMSE	Notes
Linear Regression	78.7126	108.0159	Baseline linear model
Decision Tree	36.8939	69.6571	Captures nonlinear relationships
Random Forest	34.3853	63.5656	Ensemble tree mode
XGBoost	32.8537	57.0368	Gradient-boosting model
LSTM + Random Forest	34.3814	62.8066	Combines temporal and tree-based learning
GRU + Random Forest	34.3701	62.7778	Integrates GRU temporal features with Random Forest
GRU + XGBoost	32.7027	56.8930	Combines GRU temporal features with gradient boosting
TATFF-XGBLSTM	32.6901	56.8745	LSTM temporal feature fusion with XGBoost

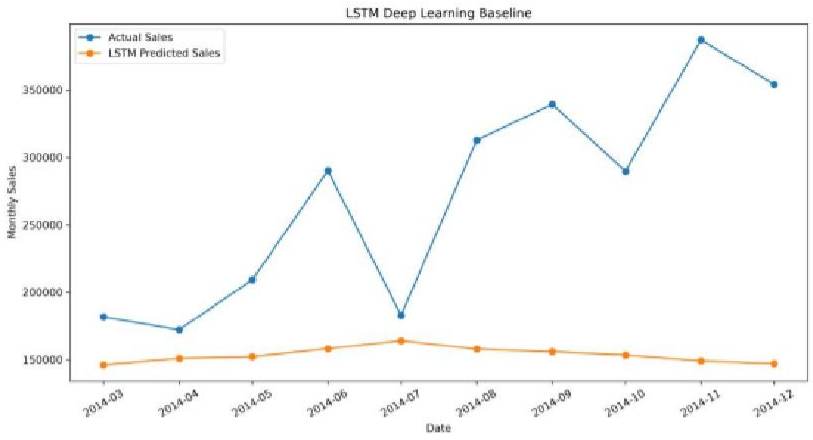
MAE and RMSE are used for measurement of the prediction error, where the lower value indicates the better prediction. The suggested TATFF-XGBLSTM model has produced MAE of 32.6901 and RMSE of 56.8745.

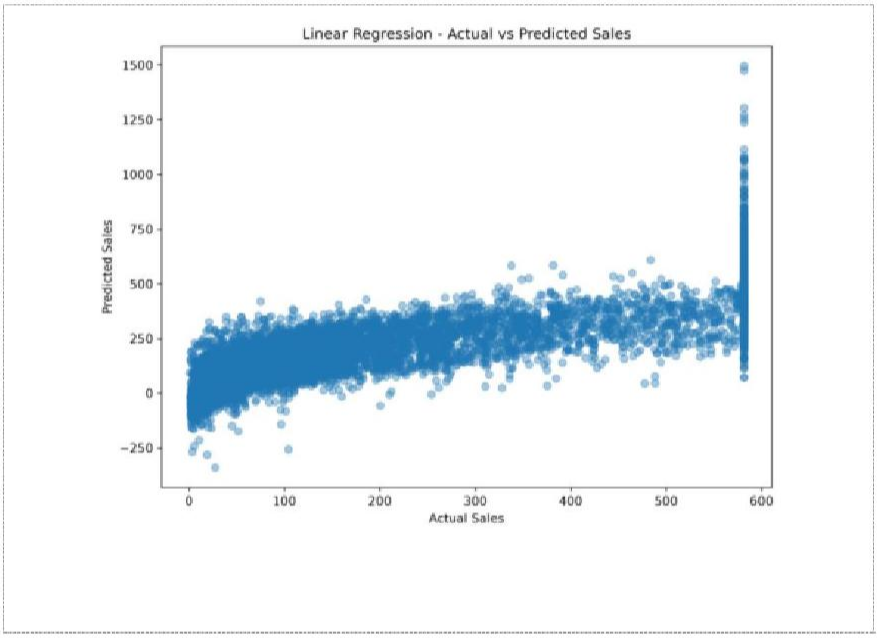



Decision Tree - Actual vs Predicted Sales
Predicted Sales
Actual Sales
0
100
200
300
400
500
600


Random Forest - Actual vs Predicted Sales
Predicted Sales
Actual Sales
0
100
200
300
400
500
600


XGBoost - Actual vs Predicted Sales
Predicted Sales
Actual Sales
0
100
200
300
400
500
600
700


LSTM Deep Learning Baseline
Actual Sales
LSTM Predicted Sales
Monthly Sales
350000
300000
250000
200000
150000
2014-03
2014-04
2014-05
2014-06
2014-07
2014-08
2014-09
2014-10
2014-11
2014-12
Date


Linear Regression - Actual vs Predicted Sales
Predicted Sales
1500
1250
1000
750
500
250
0
−250
0
100
200
300
400
500
600
Actual Sales

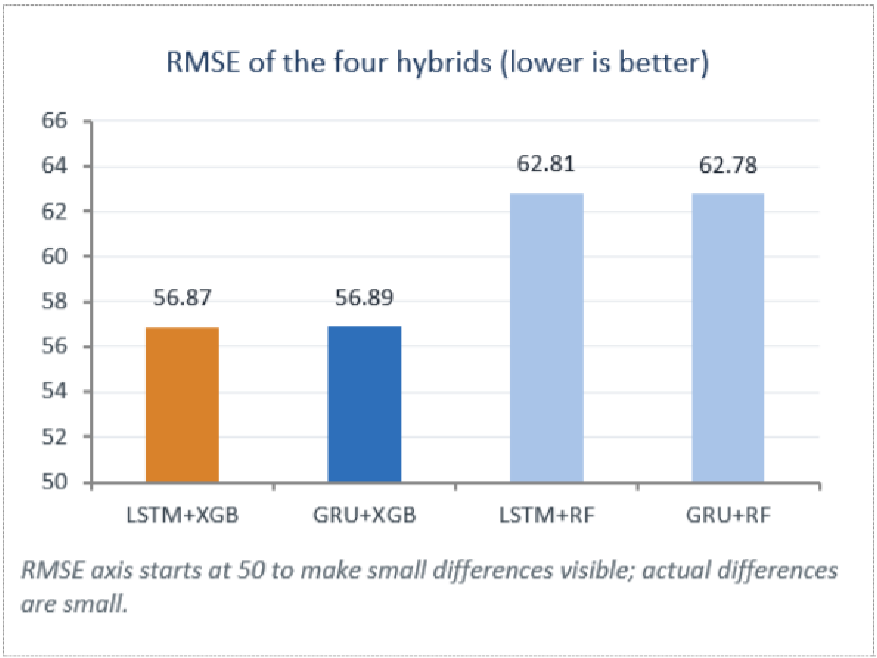


Fig. 3.

5.3 [Additional Result Sub-section, e.g., Error/Confusion Analysis]

K-Means clustering was evaluated using values of K ranging from 2 to 6.

Number of Clusters	*Silhouette Score*
2	*0.666635*
3	*0.554768*
4	*0.506601*
5	*0.355589*
6	*0.353201*

Table 3. Silhouette Score comparison for different K-Means cluster configurations.

The highest Silhouette Score is obtained when $K = 2$, with a value of ***0.666635****. The score decreases as the number of clusters increases. Therefore, the two-cluster configuration was selected for the implemented customer segmentation.*

5.4 [Additional Result Sub-section, e.g., Feature/Factor Importance]

RFM analysis was performed using Recency, Frequency, and Monetary characteristics.

Customer Value	*Customers*	*Total Sales*	*Total Profit*	*Average RFM Score*
High Value	*515*	*5,484,204*	*578,661*	*13.23*
Medium Value	*461*	*2,510,516*	*269,763*	*9.59*
Low Value	*614*	*845,582*	*79,537*	*5.01*

6. DISCUSSION

6.1 Interpretation of Findings

From the experiment, it is clear that model selection has a significant impact on the accuracy of sales prediction. Linear Regression gives a good baseline but gives relatively high error predictions. Tree-based algorithms give better results as they are able to capture nonlinear relationships and interactions between retail variables.

An XGBoost is a suitable baseline method, where the mean absolute error (MAE) score and root mean square error (RMSE) score achieved by it are 32.8537 and 57.0368, respectively. The TATFF-XGBLSTM method results in a MAE of 32.6901 and RMSE of 56.8745, which demonstrates the benefit of using temporal features for XGBoost predictions.

The increase in R^2 value from 0.6830 in Linear Regression to 0.9116 in XGBoost shows that the developed XGBoost model accounts for more variance of the observations than other models.

The proposed TATFF-XGBLSTM model was implemented to combine historical monthly sales information with transaction-level features. The LSTM component generated

temporal information from previous sales observations, and this information was incorporated into the XGBoost prediction pipeline.

The model outputs are presented through the generated prediction results, actual-versus-predicted plots, and evaluation metrics. These outputs are used to assess the performance of the hybrid model relative to the baseline models.

6.2 Comparison With Previous Studies

The results are aligned with the trend of research in the field of retail analytics, where machine learning approaches have been applied to predict sales and segment customers.

Tang analyzed the application of machine learning and RFM techniques to retail demand forecasting and customer segmentation, illustrating the benefits of predicting sales along with analyzing the customers [5].

Ayem et al. examined the use of artificial intelligence for customer profiling, segmentation, and sales prediction, emphasizing the potential of merging the customer analysis with the predictions [6].

A dynamic approach to customer segmentation has been suggested by Wang et al. through the use of LRFMS and multivariate time series clustering, emphasizing the significance of modeling customer behavior [7].

Further exploring these lines of research, the current paper presents the TATFF-XGBLSTM framework that employs LSTM-based temporal features along with XGBoost in order to make predictions based on transactions

6.3 Theoretical Implications

The findings demonstrate that retail analytics can benefit from combining several analytical perspectives rather than depending on a single model.

Sales prediction provides a predictive perspective, while clustering provides a behavioral perspective. RFM provides customer-value information, and CLV provides a financial perspective.

The integration of these approaches provides a more comprehensive representation of retail performance.

The proposed TATFF-XGBLSTM model further demonstrates the potential of integrating temporal learning with transaction-level machine learning to enhance retail sales prediction.

6.4 Practical Implications

The proposed framework can support retail organizations in several areas.

Sales prediction can assist with planning and forecasting.

Customer segmentation can help identify groups with different purchasing behavior.

RFM analysis can support customer-value identification.

CLV analysis can help identify customers with greater historical value.

Product analysis can help identify products and categories with different sales and profit characteristics.

Regional analysis can identify differences between geographic markets.

Power BI dashboards can make these results accessible through interactive business intelligence.

The Business Recommendation Engine can further convert analytical outputs into business-oriented recommendations.

The proposed hybrid model can support sales forecasting by combining historical monthly sales patterns with transaction-level attributes.

The new model suggested (TATFF-XGBLSTM) can help in retail decision-making by considering the past sales trend along with the transactions' details. The prediction of the sales made by this model can assist in making decisions regarding inventory planning and demand estimation. By applying this model along with the other techniques such as customer segmentation, RFM analysis, CLV estimation, and scenario analysis, better decisions can be made.s

6.5 Unexpected Findings

The first noteworthy thing is the discrepancy between the K-Means clusters of customers and the RFM segments.

While K-Means creates two clusters using customer-level behavioral variables, RFM creates three segments of customers according to their value.

This illustrates that various approaches allow capturing various aspects of customer behavior. Consequently, these results should not be considered to mean the same classification of customers but rather to complement each other.

The second thing is that Office Supplies earned more profit than Technology although Technology made more sales.

The proposed hybrid model also showed a modest improvement over the standalone XGBoost baseline in the evaluated experiment. This indicates that the LSTM-generated temporal feature may provide additional information for transaction-level sales prediction.

7. LIMITATIONS

- Evaluation of the regression models involved a one-time hold-out test using an 80:20 ratio instead of cross-validation.
- The stated predictions for this task are based on the Global Superstore data set and will not necessarily apply to other retail data sets.

- The data set is from past transactions in the retail business and does not necessarily reflect present conditions.
- Present implementation of CLV gives the customer's historical value but not a probabilistic future lifetime model.
- The sensitivity analysis is model-based and hence does not demonstrate causality.
- In this way, the TATFF-XGBLSTM model can be useful for retail decision making because it incorporates both historic monthly sales patterns as well as transaction-level attributes.
- The forecasting ability of the model can be helpful in managing inventories and calculating demand. In combination with RFM modeling, customer segmentations, CLV calculations, and scenario building, the results can provide further insight.

8. CONCLUSION AND FUTURE WORK

In this research work, a Retail Sales Analytics and Customer Intelligence Framework that uses Data Preprocessing, Exploratory Data Analysis, Feature Engineering, Machine Learning, Customer Segmentation, RFM Analysis, Customer Lifetime Value, SQL, Scenario Analysis, and Power BI was developed.

Four different regression models were tested for sales forecasting purposes. In particular, XGBoost scored an MAE of 32.8537, RMSE of 57.0368, and R^2 of 0.9116 on the built test dataset. As for K-Means Clustering, its best Silhouette Score on two clusters is 0.666635.

Additionally, the proposed TATFF-XGBLSTM hybrid model integrates LSTM-generated temporal features with XGBoost-based transaction-level prediction. In the evaluated experiment, the hybrid model achieved an MAE of 32.6901 and an RMSE of 56.8745, compared with 32.8537 and 57.0368, respectively, for the previously reported XGBoost baseline. These results suggest a potential improvement from temporal feature integration, although further validation is required to establish the generalizability of the findings.

The customer intelligence approach delivers complementary data through K-Means clustering, RFM analysis, and CLV analysis. Moreover, Power BI dashboards deliver an interactive environment for studying sales, products, customers, CLV, scenarios, and business-related information.

The novel idea of the Business Recommendation Engine is the main innovation of the research work

Future Work:

Future work will focus on the following improvements:

1. *Implementing repeated k-fold and temporal cross-validation for more robust model evaluation.*
2. *Testing the framework on additional Superstore benchmark datasets.*
3. *Integrating SHAP and LIME for explainable machine learning.*
4. *Developing a fully automated Business Recommendation Engine.*
5. *Implementing advanced Customer Lifetime Value prediction.*
6. *Adding demand and inventory forecasting.*
7. *Developing a Streamlit-based deployment interface.*
8. *Investigating real-time retail data integration*

DECLARATIONS

Funding

No funding was received for the completion of this research project from any external source. It was done as an academic project using freely available retail data.

Conflict of Interest

The authors declare that there is no conflict of interest related to this research.

Ethics Statement

study is based on publicly available sales data and does not require any interaction with individuals. No personally identifiable information about customers was knowingly used in the study.

Data Availability

The research is based on Global Superstore data collected from a publicly accessible source at Kaggle. The project roadmap also mentions publicly accessible benchmark datasets of Superstores from Kaggle and Tableau.

Author Contributions

J. Madhura: *Statistical analysis, visualization, customer analytics, and dashboard development.*

R. Lokesh: *Machine-learning implementation, model evaluation, customer segmentation, and scenario analysis.*

G. Charan: *SQL/database integration, Power BI integration, documentation, and presentation.*

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